OPENSOURCESDRLAB DOCUMENTATION

USRP-B210-Artix-7-AD9361 Schematic

Product	USRP-B210-Artix-7-AD9361
Document	OSSDRLAB-USRP-B210-Artix-7-AD9361-SCH
Version	1.0
Date	2026-11-12

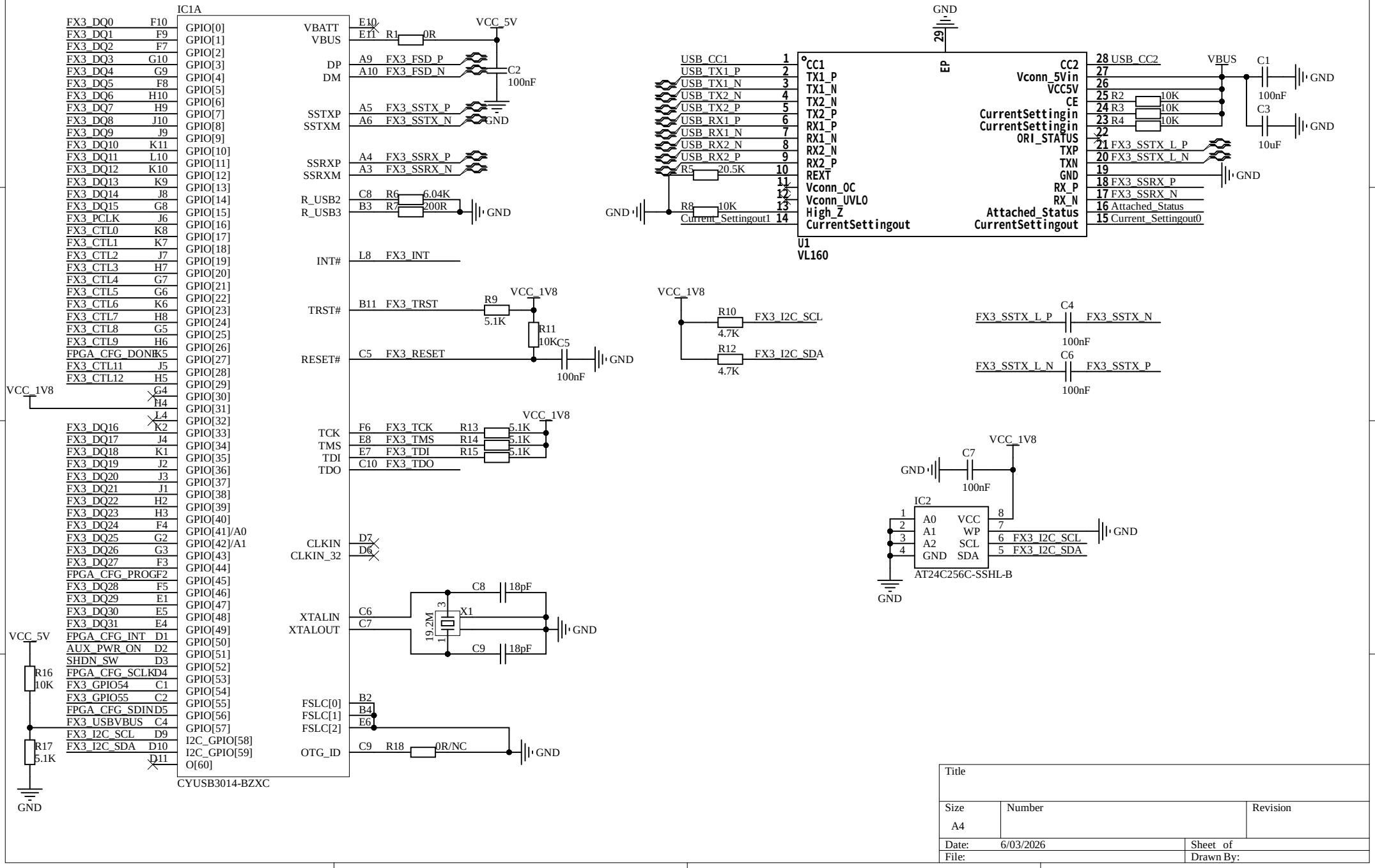
Circuit schematic of the OpenSourceSDRLab USRP B210 (Artix-7), a USRP B210-compatible SDR with an Analog Devices AD9361 and a Xilinx Artix-7 XC7A75T or XC7A200T FPGA. Exported from Altium Designer (A4); sheet 13 is blank. The short Chinese notes on sheets 6 and 7 mark pins that may be swapped within an FPGA bank; the note on sheet 12 says that there is no onboard GPSDO (external reference inputs only).

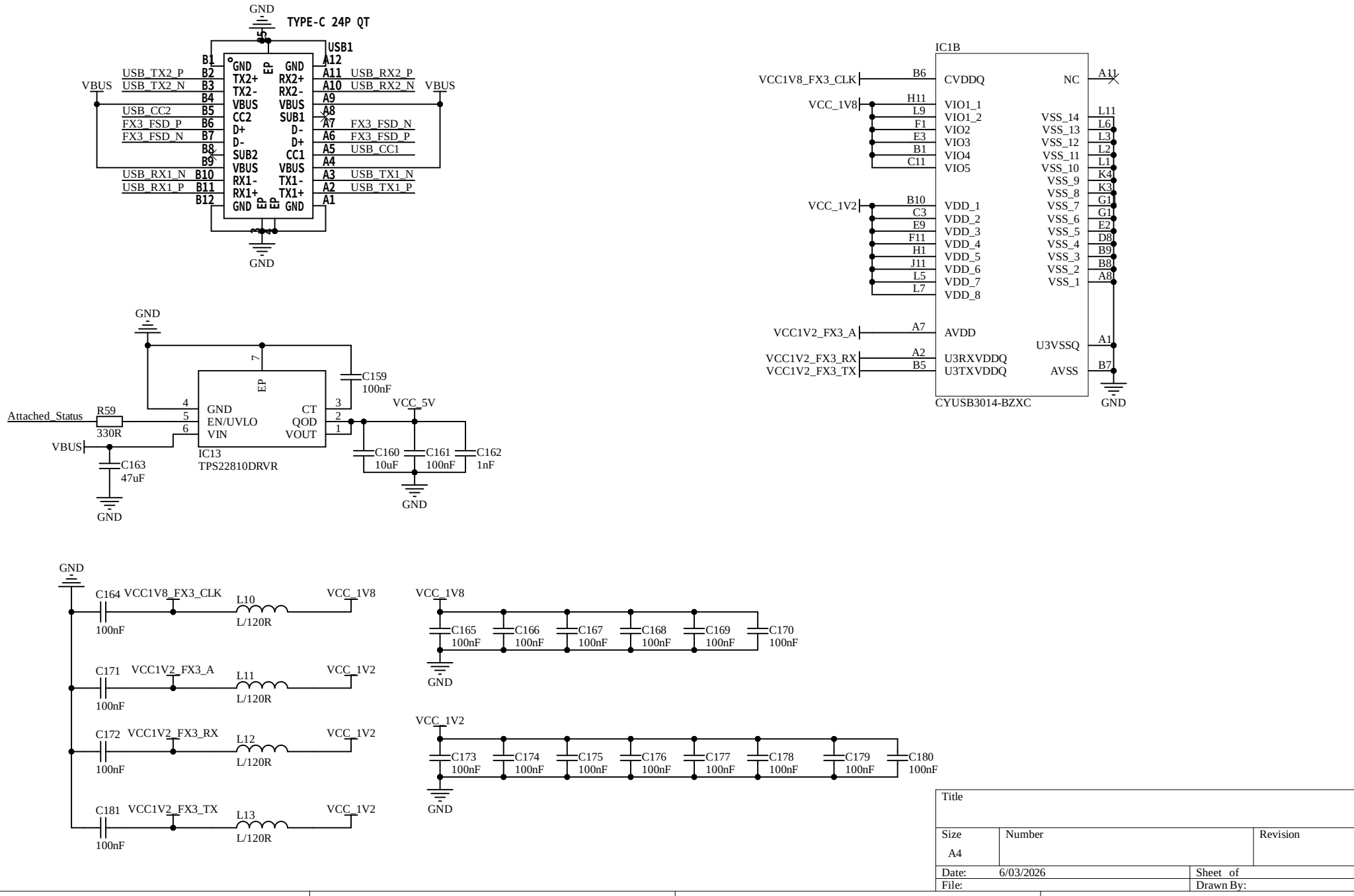
Sheet	Content
1–2	USB 3.0: CYUSB3014 (FX3) controller, GPIF II bus, Type-C connector and FX3 supply
3	Power: MP2145 step-down converters, TPS7A9101 low-noise LDOs
4–5	FPGA configuration, power pins and decoupling
6–7	FPGA I/O banks: AD9361 interface, FX3 GPIF II, PLL control, GPIO
8	AD9361 RF transceiver
9–10	RF front ends, channel 1 (TRX1, RX1) and channel 2 (TRX2, RX2)
11	Status LEDs, MX1.25 GPIO connector, FPGA and FX3 JTAG
12	Clocking: ADF4001 PLL, external 10 MHz and PPS inputs (no onboard GPSDO)

Product page: https://opensourcesdrlab.com/products/opensourcesdrlab-usrp-b210-ad9361-and-xilinx-xc7a75t-or-xc7a200t

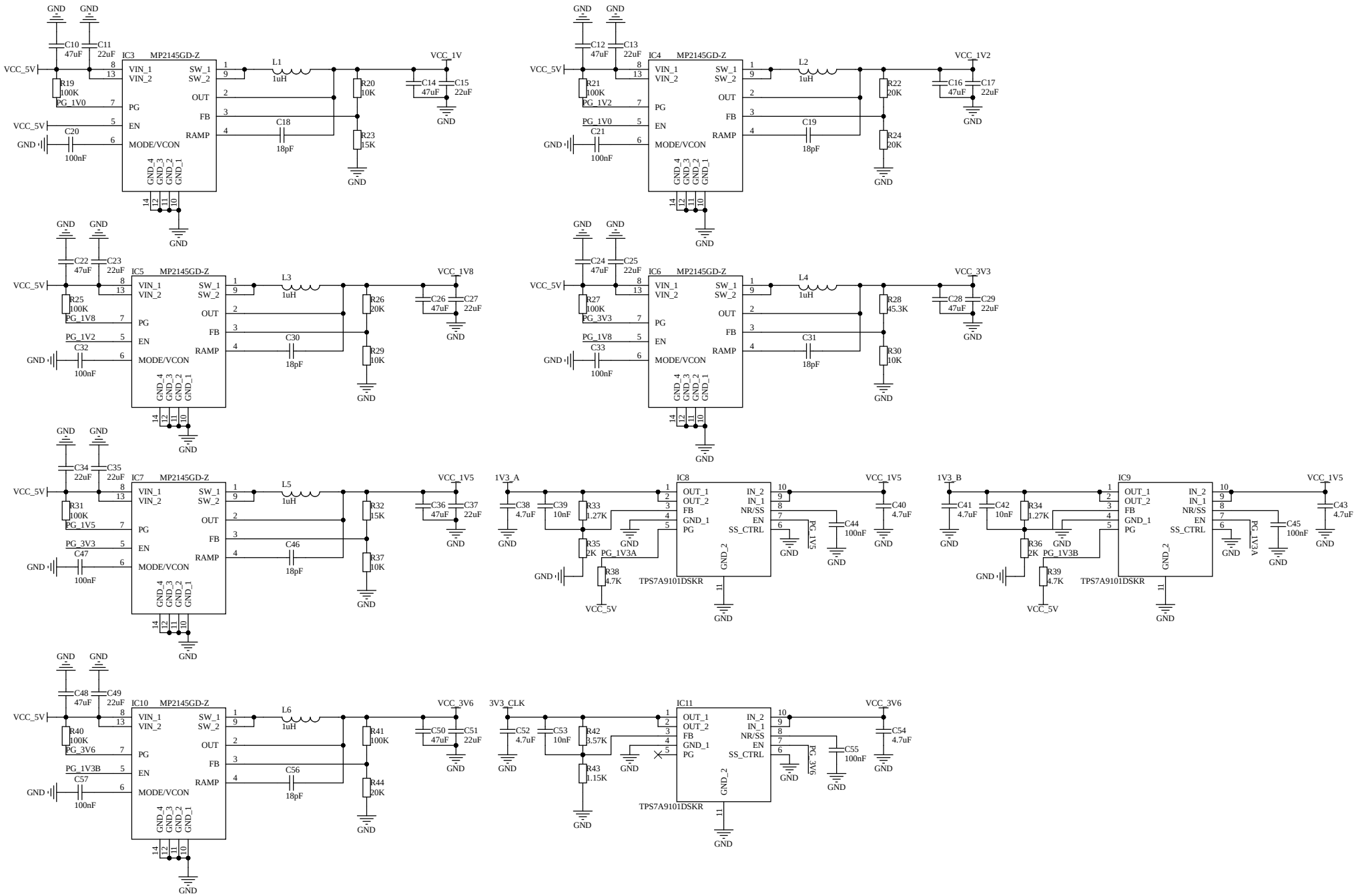
Questions about the hardware: support@opensourcesdrlab.com

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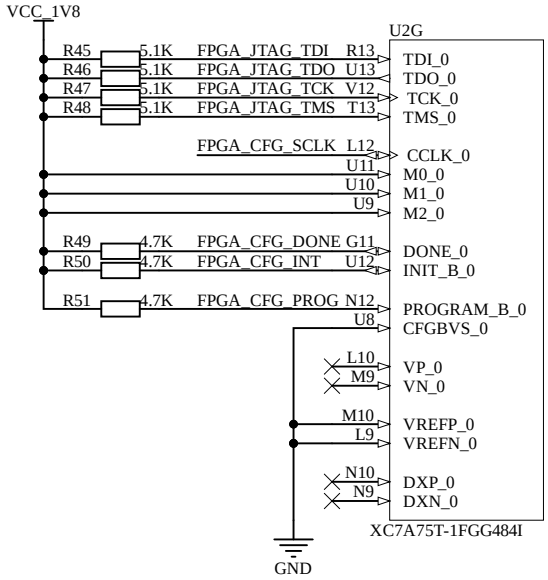

IC1A
CYUSB3014-BZXC
FX3_DQ0 F10
FX3_DQ1 F9
FX3_DQ2 F7
FX3_DQ3 G10
FX3_DQ4 G9
FX3_DQ5 F8
FX3_DQ6 H10
FX3_DQ7 H9
FX3_DQ8 J10
FX3_DQ9 J9
FX3_DQ10 K11
FX3_DQ11 L10
FX3_DQ12 K10
FX3_DQ13 K9
FX3_DQ14 J8
FX3_DQ15 G8
FX3_PCLK J6
FX3_CTL0 K8
FX3_CTL1 K7
FX3_CTL2 J7
FX3_CTL3 H7
FX3_CTL4 G7
FX3_CTL5 G6
FX3_CTL6 K6
FX3_CTL7 H8
FX3_CTL8 G5
FX3_CTL9 H6
FPGA_CFG_DONE K5
FX3_CTL11 J5
FX3_CTL12 H5
VCC_1V8
G4
H4
L4
FX3_DQ16 K2
FX3_DQ17 J4
FX3_DQ18 K1
FX3_DQ19 J2
FX3_DQ20 J3
FX3_DQ21 J1
FX3_DQ22 H2
FX3_DQ23 H3
FX3_DQ24 F4
FX3_DQ25 G2
FX3_DQ26 G3
FX3_DQ27 F3
FPGA_CFG_PROG F2
FX3_DQ28 F5
FX3_DQ29 E1
FX3_DQ30 E5
FX3_DQ31 E4
VCC_5V
FPGA_CFG_INT D1
AUX_PWR_ON D2
SHDN_SW D3
FPGA_CFG_SCLK D4
FX3_GPIO54 C1
FX3_GPIO55 C2
FPGA_CFG_SDIN D5
FX3_USBVBUS C4
FX3_I2C_SCL D9
FX3_I2C_SDA D10
D11
R16 10K
R17 5.1K
GND
GPIO[0]
GPIO[1]
GPIO[2]
GPIO[3]
GPIO[4]
GPIO[5]
GPIO[6]
GPIO[7]
GPIO[8]
GPIO[9]
GPIO[10]
GPIO[11]
GPIO[12]
GPIO[13]
GPIO[14]
GPIO[15]
GPIO[16]
GPIO[17]
GPIO[18]
GPIO[19]
GPIO[20]
GPIO[21]
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GPIO[28]
GPIO[29]
GPIO[30]
GPIO[31]
GPIO[32]
GPIO[33]
GPIO[34]
GPIO[35]
GPIO[36]
GPIO[37]
GPIO[38]
GPIO[39]
GPIO[40]
GPIO[41]/A0
GPIO[42]/A1
GPIO[43]
GPIO[44]
GPIO[45]
GPIO[46]
GPIO[47]
GPIO[48]
GPIO[49]
GPIO[50]
GPIO[51]
GPIO[52]
GPIO[53]
GPIO[54]
GPIO[55]
GPIO[56]
GPIO[57]
I2C_GPIO[58]
I2C_GPIO[59]
O[60]
VBATT
VBUS
DP
DM
SSTXP
SSTXM
SSRXP
SSRXM
R_USB2
R_USB3
INT#
TRST#
RESET#
TCK
TMS
TDI
TDO
CLKIN
CLKIN_32
XTALIN
XTALOUT
FSLC[0]
FSLC[1]
FSLC[2]
OTG_ID
E10
E11 R1 0R
VCC_5V
A9 FX3_FSD_P
A10 FX3_FSD_N
C2 100nF
A5 FX3_SSTX_P
A6 FX3_SSTX_N
GND
A4 FX3_SSRX_P
A3 FX3_SSRX_N
C8 R6 6.04K
B3 R7 200R
GND
L8 FX3_INT
B11 FX3_TRST
R9 5.1K
VCC_1V8
R11 10K
C5 100nF
GND
C5 FX3_RESET
F6 FX3_TCK R13 5.1K
E8 FX3_TMS R14 5.1K
E7 FX3_TDI R15 5.1K
VCC_1V8
C10 FX3_TDO
D7
D6
C8 18pF
C6
C7
19.2M
X1
C9 18pF
GND
B2
B4
E6
C9 R18 0R/NC
GND
U1
VL160
EP
29
GND
USB_CC1 1
USB_TX1_P 2
USB_TX1_N 3
USB_TX2_N 4
USB_TX2_P 5
USB_RX1_P 6
USB_RX1_N 7
USB_RX2_N 8
USB_RX2_P 9
R5 20.5K 10
11
12
13
R8 10K 14
Current_Settingout1
GND
CC1
TX1_P
TX1_N
TX2_N
TX2_P
RX1_P
RX1_N
RX2_N
RX2_P
REXT
Vconn_OC
Vconn_UVLO
High_Z
CurrentSettingout
CC2
Vconn_5Vin
VCC5V
CE
CurrentSettingin
CurrentSettingin
ORI_STATUS
TXP
TXN
GND
RX_P
RX_N
Attached_Status
CurrentSettingout
28 USB_CC2
27
26
25 R2 10K
24 R3 10K
23 R4 10K
22
21 FX3_SSTX_L_P
20 FX3_SSTX_L_N
19
18 FX3_SSRX_P
17 FX3_SSRX_N
16 Attached_Status
15 Current_Settingout0
VBUS
C1 100nF
GND
C3 10uF
GND
GND
VCC_1V8
R10 4.7K
FX3_I2C_SCL
R12 4.7K
FX3_I2C_SDA
FX3_SSTX_L_P
C4 100nF
FX3_SSTX_N
FX3_SSTX_L_N
C6 100nF
FX3_SSTX_P
VCC_1V8
GND
C7 100nF
IC2
A0
A1
A2
GND
VCC
WP
SCL
SDA
8
7
6 FX3_I2C_SCL
5 FX3_I2C_SDA
GND
AT24C256C-SSHL-B
GND
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Sheet of
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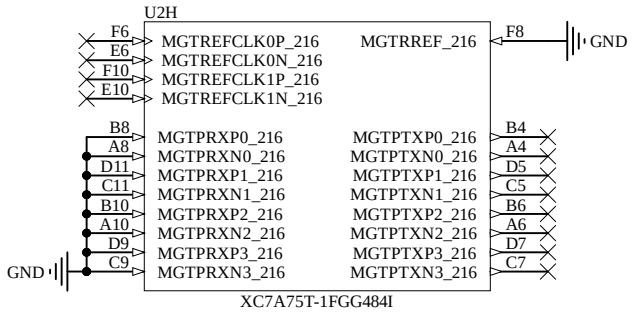


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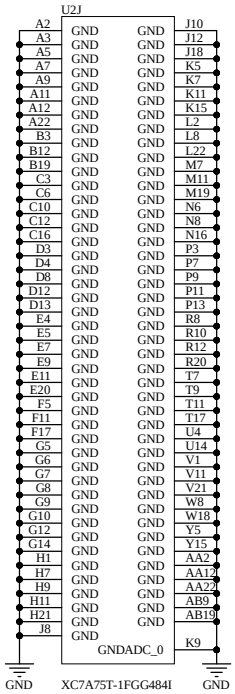


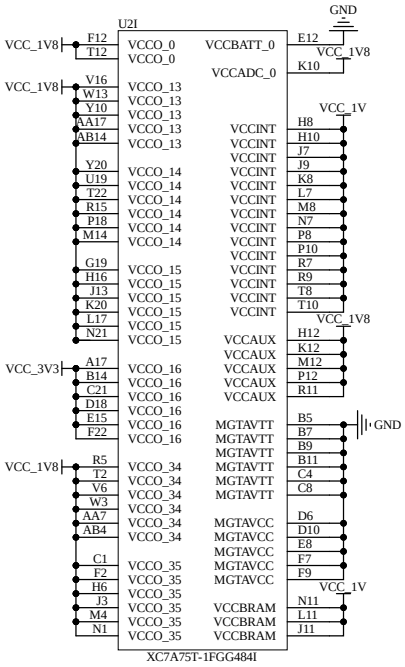
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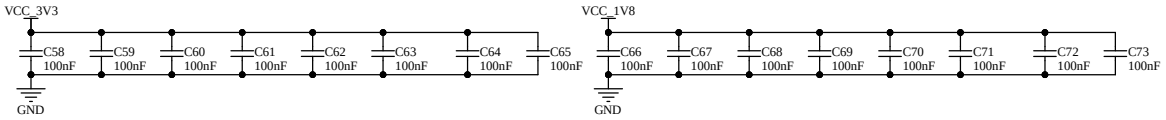


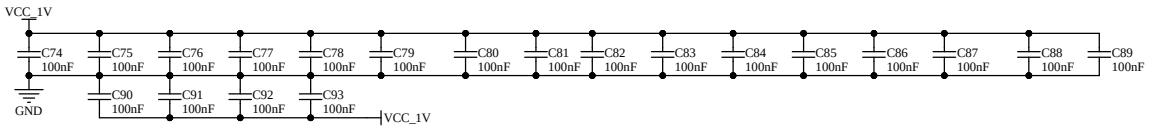


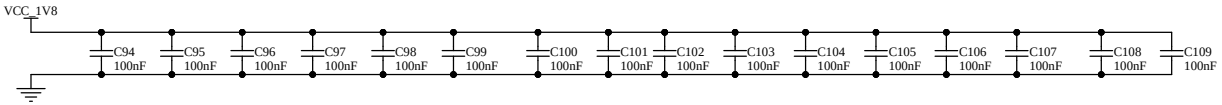
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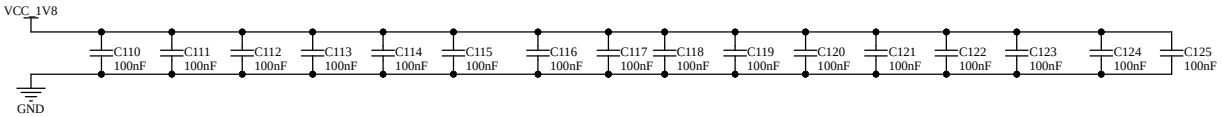


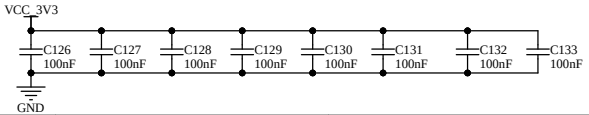




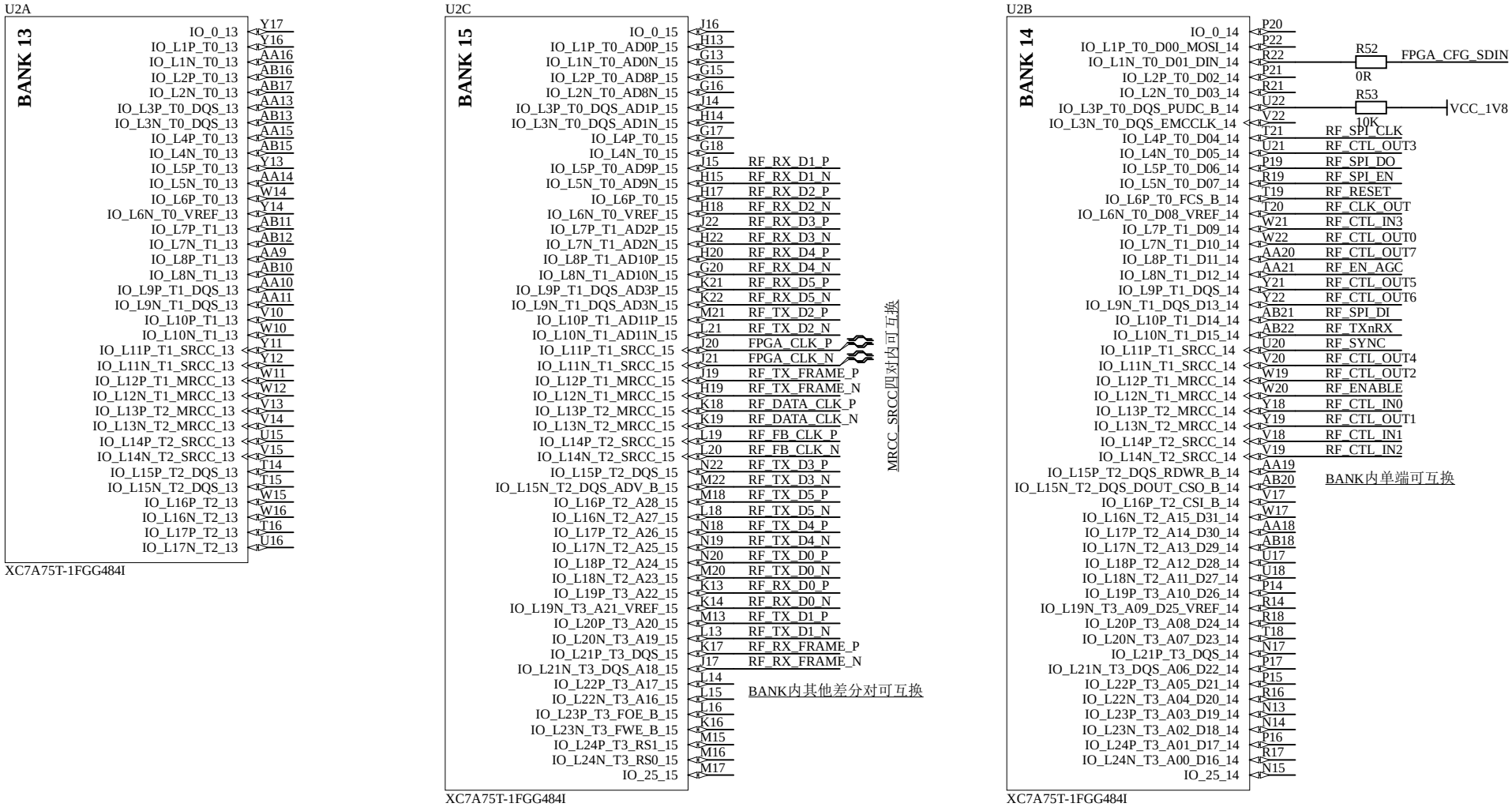


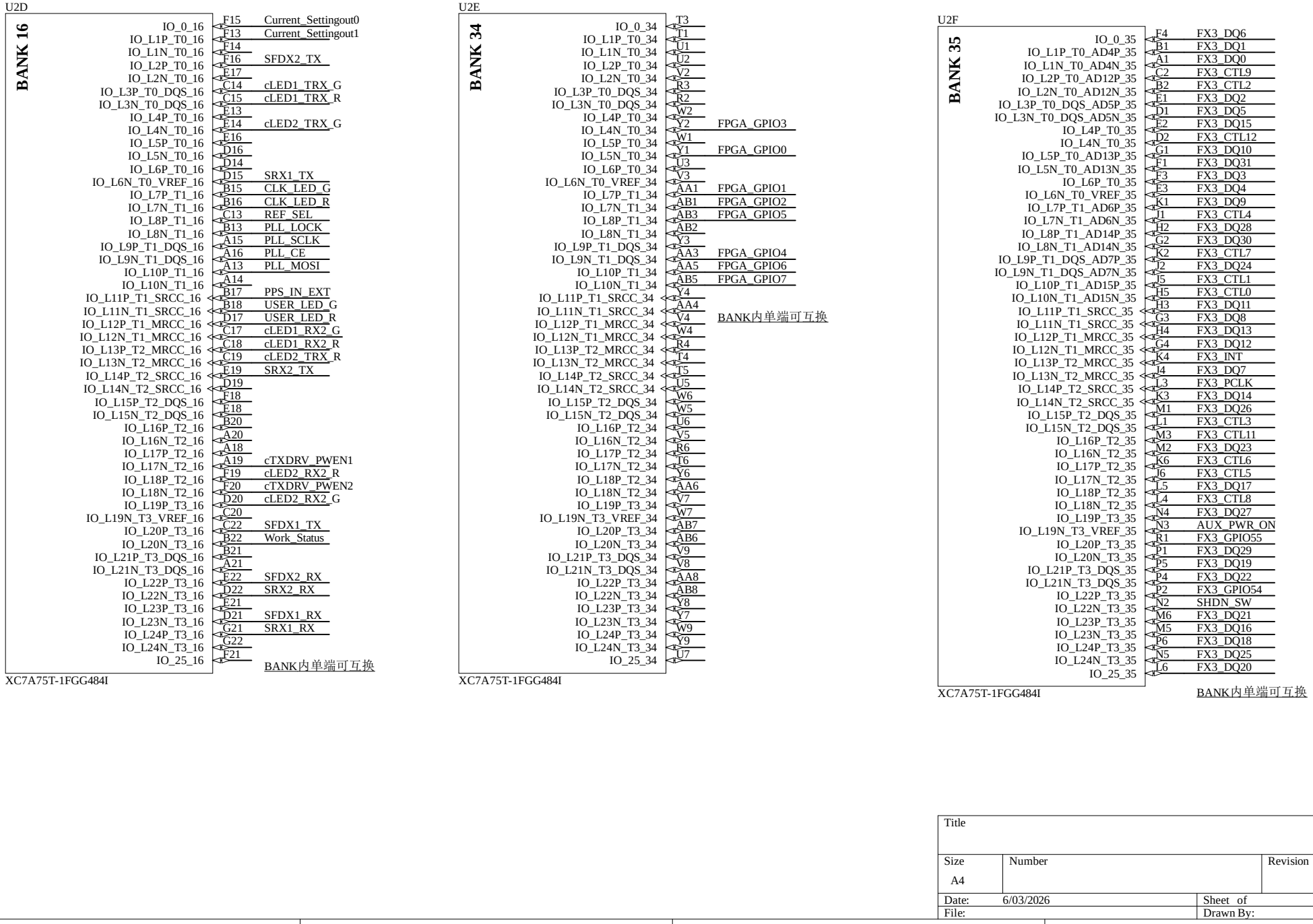


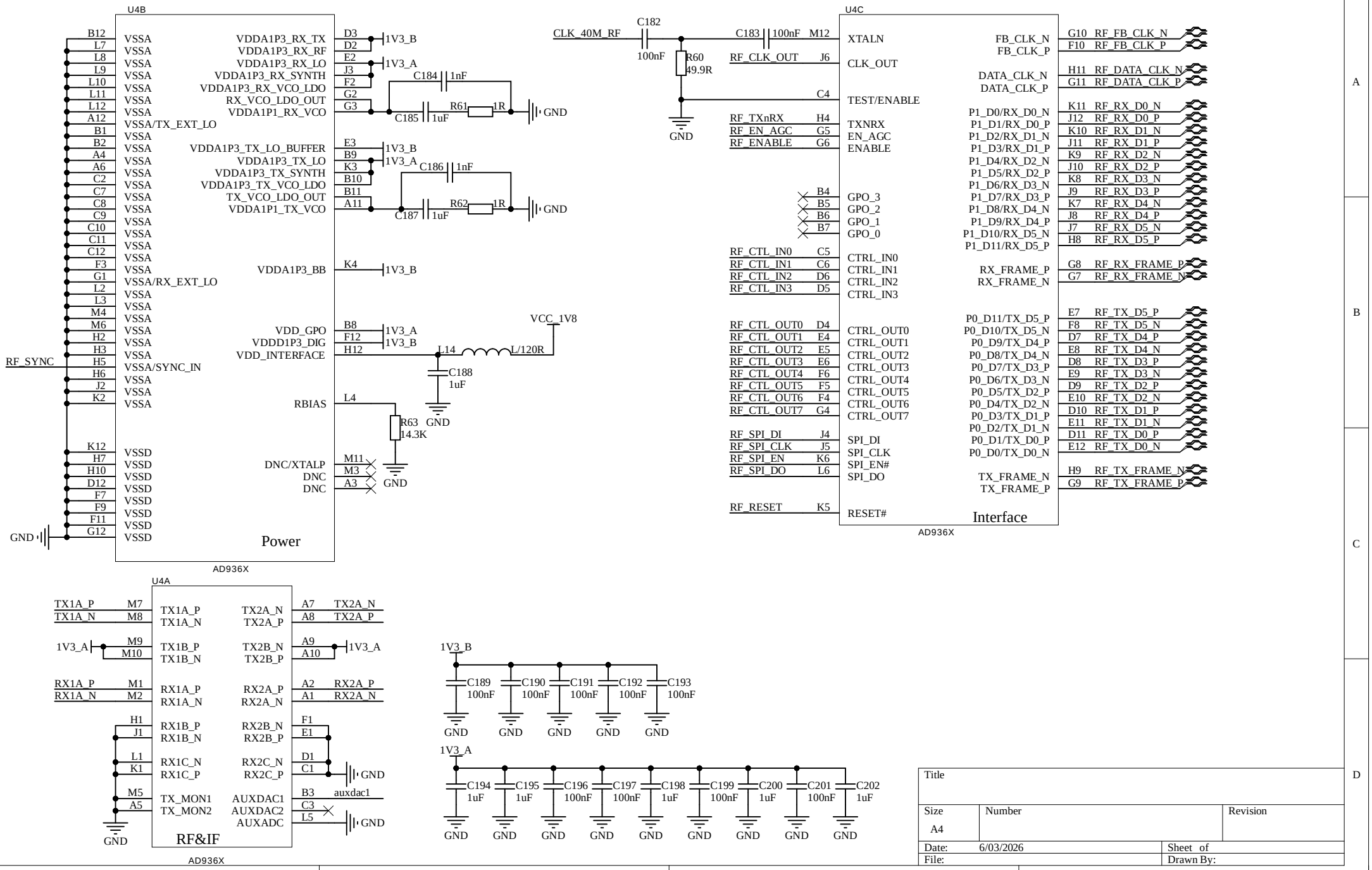


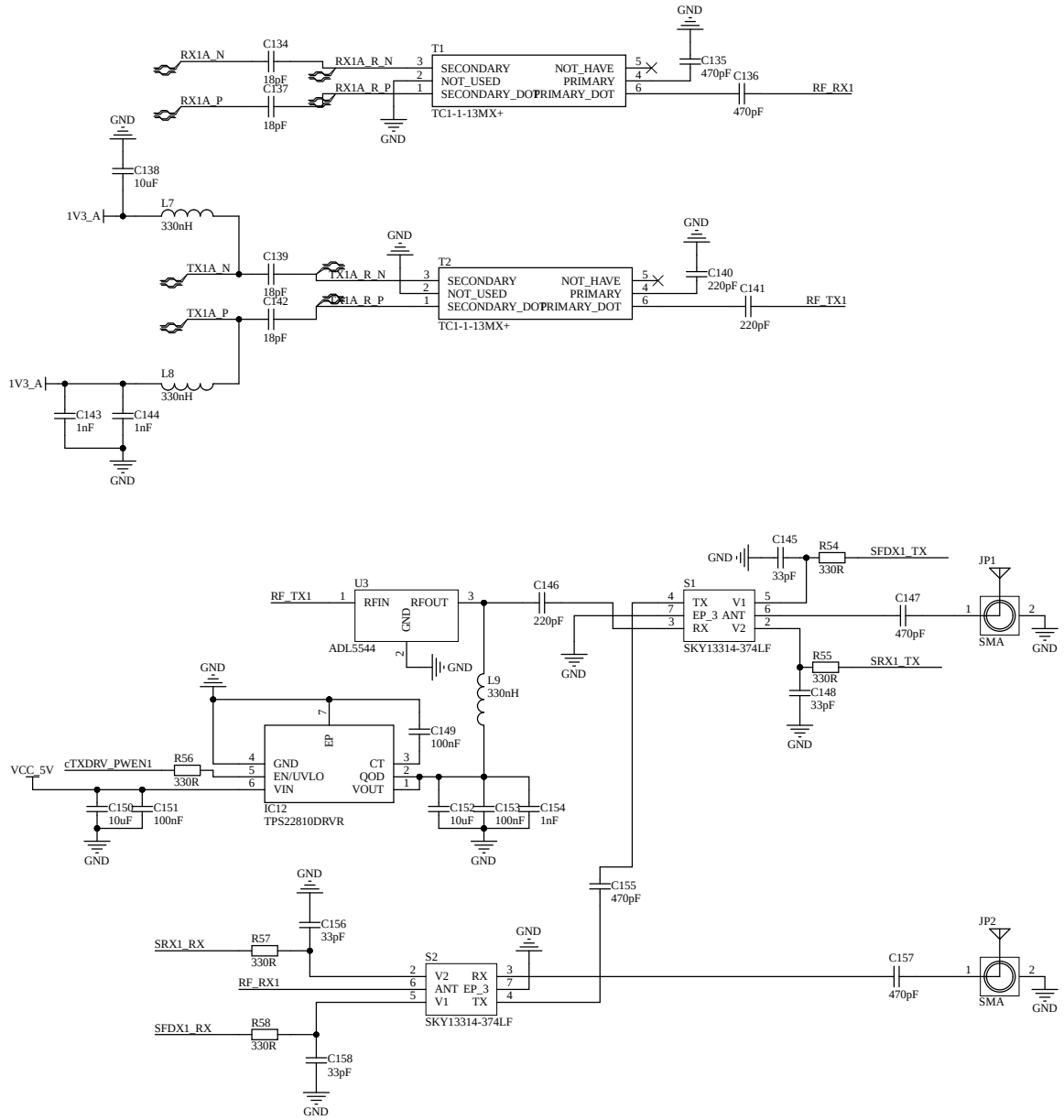


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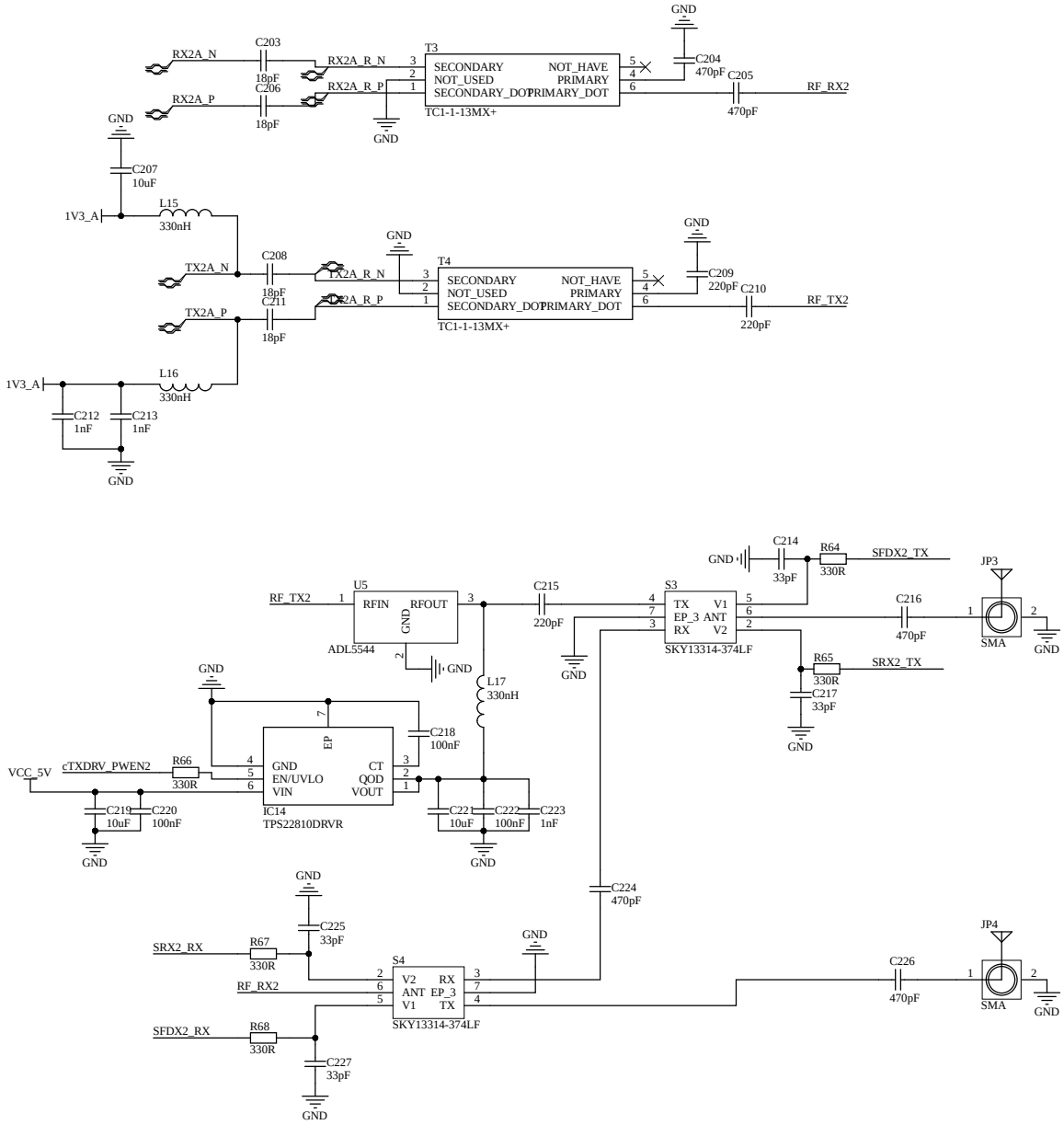

U2A
BANK 13
IO_0_13 Y17
IO_L1P_T0_13 Y16
IO_L1N_T0_13 AA16
IO_L2P_T0_13 AB16
IO_L2N_T0_13 AB17
IO_L3P_T0_DQS_13 AA13
IO_L3N_T0_DQS_13 AB13
IO_L4P_T0_13 AA15
IO_L4N_T0_13 AB15
IO_L5P_T0_13 Y13
IO_L5N_T0_13 AA14
IO_L6P_T0_13 W14
IO_L6N_T0_VREF_13 Y14
IO_L7P_T1_13 AB11
IO_L7N_T1_13 AB12
IO_L8P_T1_13 AA9
IO_L8N_T1_13 AB10
IO_L9P_T1_DQS_13 AA10
IO_L9N_T1_DQS_13 AA11
IO_L10P_T1_13 V10
IO_L10N_T1_13 W10
IO_L11P_T1_SRCC_13 Y11
IO_L11N_T1_SRCC_13 Y12
IO_L12P_T1_MRCC_13 W11
IO_L12N_T1_MRCC_13 W12
IO_L13P_T2_MRCC_13 V13
IO_L13N_T2_MRCC_13 V14
IO_L14P_T2_SRCC_13 U15
IO_L14N_T2_SRCC_13 V15
IO_L15P_T2_DQS_13 T14
IO_L15N_T2_DQS_13 T15
IO_L16P_T2_13 W15
IO_L16N_T2_13 W16
IO_L17P_T2_13 T16
IO_L17N_T2_13 U16
XC7A75T-1FGG484I
U2C
BANK 15
IO_0_15 J16
IO_L1P_T0_AD0P_15 H13
IO_L1N_T0_AD0N_15 G13
IO_L2P_T0_AD8P_15 G15
IO_L2N_T0_AD8N_15 G16
IO_L3P_T0_DQS_AD1P_15 J14
IO_L3N_T0_DQS_AD1N_15 H14
IO_L4P_T0_15 G17
IO_L4N_T0_15 G18
IO_L5P_T0_AD9P_15 J15 RF_RX_D1_P
IO_L5N_T0_AD9N_15 H15 RF_RX_D1_N
IO_L6P_T0_15 H17 RF_RX_D2_P
IO_L6N_T0_VREF_15 H18 RF_RX_D2_N
IO_L7P_T1_AD2P_15 J22 RF_RX_D3_P
IO_L7N_T1_AD2N_15 H22 RF_RX_D3_N
IO_L8P_T1_AD10P_15 H20 RF_RX_D4_P
IO_L8N_T1_AD10N_15 G20 RF_RX_D4_N
IO_L9P_T1_DQS_AD3P_15 K21 RF_RX_D5_P
IO_L9N_T1_DQS_AD3N_15 K22 RF_RX_D5_N
IO_L10P_T1_AD11P_15 M21 RF_TX_D2_P
IO_L10N_T1_AD11N_15 L21 RF_TX_D2_N
IO_L11P_T1_SRCC_15 J20 FPGA_CLK_P
IO_L11N_T1_SRCC_15 J21 FPGA_CLK_N
IO_L12P_T1_MRCC_15 J19 RF_TX_FRAME_P
IO_L12N_T1_MRCC_15 H19 RF_TX_FRAME_N
IO_L13P_T2_MRCC_15 K18 RF_DATA_CLK_P
IO_L13N_T2_MRCC_15 K19 RF_DATA_CLK_N
IO_L14P_T2_SRCC_15 L19 RF_FB_CLK_P
IO_L14N_T2_SRCC_15 L20 RF_FB_CLK_N
IO_L15P_T2_DQS_15 N22 RF_TX_D3_P
IO_L15N_T2_DQS_ADV_B_15 M22 RF_TX_D3_N
IO_L16P_T2_A28_15 M18 RF_TX_D5_P
IO_L16N_T2_A27_15 L18 RF_TX_D5_N
IO_L17P_T2_A26_15 N18 RF_TX_D4_P
IO_L17N_T2_A25_15 N19 RF_TX_D4_N
IO_L18P_T2_A24_15 N20 RF_TX_D0_P
IO_L18N_T2_A23_15 M20 RF_TX_D0_N
IO_L19P_T3_A22_15 K13 RF_RX_D0_P
IO_L19N_T3_A21_VREF_15 K14 RF_RX_D0_N
IO_L20P_T3_A20_15 M13 RF_TX_D1_P
IO_L20N_T3_A19_15 L13 RF_TX_D1_N
IO_L21P_T3_DQS_15 K17 RF_RX_FRAME_P
IO_L21N_T3_DQS_A18_15 J17 RF_RX_FRAME_N
IO_L22P_T3_A17_15 L14
IO_L22N_T3_A16_15 L15
IO_L23P_T3_FOE_B_15 L16
IO_L23N_T3_FWE_B_15 K16
IO_L24P_T3_RS1_15 M15
IO_L24N_T3_RS0_15 M16
IO_25_15 M17
MRCC_SRCC四对内可互换
BANK内其他差分对可互换
XC7A75T-1FGG484I
U2B
BANK 14
IO_0_14 P20
IO_L1P_T0_D00_MOSI_14 P22
IO_L1N_T0_D01_DIN_14 R22
IO_L2P_T0_D02_14 P21
IO_L2N_T0_D03_14 R21
IO_L3P_T0_DQS_PUDC_B_14 U22
IO_L3N_T0_DQS_EMCCLK_14 Y22
IO_L4P_T0_D04_14 T21 RF_SPI_CLK
IO_L4N_T0_D05_14 U21 RF_CTL_OUT3
IO_L5P_T0_D06_14 P19 RF_SPI_DO
IO_L5N_T0_D07_14 R19 RF_SPI_EN
IO_L6P_T0_FCS_B_14 T19 RF_RESET
IO_L6N_T0_D08_VREF_14 T20 RF_CLK_OUT
IO_L7P_T1_D09_14 W21 RF_CTL_IN3
IO_L7N_T1_D10_14 W22 RF_CTL_OUT0
IO_L8P_T1_D11_14 AA20 RF_CTL_OUT7
IO_L8N_T1_D12_14 AA21 RF_EN_AGC
IO_L9P_T1_DQS_14 Y21 RF_CTL_OUT5
IO_L9N_T1_DQS_D13_14 Y22 RF_CTL_OUT6
IO_L10P_T1_D14_14 AB21 RF_SPI_DI
IO_L10N_T1_D15_14 AB22 RF_TXnRX
IO_L11P_T1_SRCC_14 U20 RF_SYNC
IO_L11N_T1_SRCC_14 V20 RF_CTL_OUT4
IO_L12P_T1_MRCC_14 W19 RF_CTL_OUT2
IO_L12N_T1_MRCC_14 W20 RF_ENABLE
IO_L13P_T2_MRCC_14 Y18 RF_CTL_IN0
IO_L13N_T2_MRCC_14 Y19 RF_CTL_OUT1
IO_L14P_T2_SRCC_14 V18 RF_CTL_IN1
IO_L14N_T2_SRCC_14 V19 RF_CTL_IN2
IO_L15P_T2_DQS_RDWR_B_14 AA19
IO_L15N_T2_DQS_DOUT_CSO_B_14 AB20
IO_L16P_T2_CSI_B_14 V17
IO_L16N_T2_A15_D31_14 W17
IO_L17P_T2_A14_D30_14 AA18
IO_L17N_T2_A13_D29_14 AB18
IO_L18P_T2_A12_D28_14 U17
IO_L18N_T2_A11_D27_14 U18
IO_L19P_T3_A10_D26_14 P14
IO_L19N_T3_A09_D25_VREF_14 R14
IO_L20P_T3_A08_D24_14 R18
IO_L20N_T3_A07_D23_14 T18
IO_L21P_T3_DQS_14 N17
IO_L21N_T3_DQS_A06_D22_14 P17
IO_L22P_T3_A05_D21_14 P15
IO_L22N_T3_A04_D20_14 R16
IO_L23P_T3_A03_D19_14 N13
IO_L23N_T3_A02_D18_14 N14
IO_L24P_T3_A01_D17_14 P16
IO_L24N_T3_A00_D16_14 R17
IO_25_14 N15
R52
0R
FPGA_CFG_SDIN
R53
10K
VCC_1V8
BANK内单端可互换
XC7A75T-1FGG484I
Title
Size
A4
Number
Revision
Date: 6/03/2026
Sheet of
File:
Drawn By:


U2D
BANK 16
IO_0_16 F15 Current_Settingout0
IO_L1P_T0_16 E13 Current_Settingout1
IO_L1N_T0_16 F14
IO_L2P_T0_16 F16 SFDX2_TX
IO_L2N_T0_16 E17
IO_L3P_T0_DQS_16 C14 cLED1_TRX_G
IO_L3N_T0_DQS_16 C15 cLED1_TRX_R
IO_L4P_T0_16 E13
IO_L4N_T0_16 E14 cLED2_TRX_G
IO_L5P_T0_16 E16
IO_L5N_T0_16 D16
IO_L6P_T0_16 D14
IO_L6N_T0_VREF_16 D15 SRX1_TX
IO_L7P_T1_16 B15 CLK_LED_G
IO_L7N_T1_16 B16 CLK_LED_R
IO_L8P_T1_16 C13 REF_SEL
IO_L8N_T1_16 B13 PLL_LOCK
IO_L9P_T1_DQS_16 A15 PLL_SCLK
IO_L9N_T1_DQS_16 A16 PLL_CE
IO_L10P_T1_16 A13 PLL_MOSI
IO_L10N_T1_16 A14
IO_L11P_T1_SRCC_16 B17 PPS_IN_EXT
IO_L11N_T1_SRCC_16 B18 USER_LED_G
IO_L12P_T1_MRCC_16 D17 USER_LED_R
IO_L12N_T1_MRCC_16 C17 cLED1_RX2_G
IO_L13P_T2_MRCC_16 C18 cLED1_RX2_R
IO_L13N_T2_MRCC_16 C19 cLED2_TRX_R
IO_L14P_T2_SRCC_16 E19 SRX2_TX
IO_L14N_T2_SRCC_16 D19
IO_L15P_T2_DQS_16 F18
IO_L15N_T2_DQS_16 E18
IO_L16P_T2_16 B20
IO_L16N_T2_16 A20
IO_L17P_T2_16 A18
IO_L17N_T2_16 A19 cTXDRV_PWEN1
IO_L18P_T2_16 F19 cLED2_RX2_R
IO_L18N_T2_16 F20 cTXDRV_PWEN2
IO_L19P_T3_16 D20 cLED2_RX2_G
IO_L19N_T3_VREF_16 C20
IO_L20P_T3_16 C22 SFDX1_TX
IO_L20N_T3_16 B22 Work_Status
IO_L21P_T3_DQS_16 B21
IO_L21N_T3_DQS_16 A21
IO_L22P_T3_16 E22 SFDX2_RX
IO_L22N_T3_16 D22 SRX2_RX
IO_L23P_T3_16 E21
IO_L23N_T3_16 D21 SFDX1_RX
IO_L24P_T3_16 G21 SRX1_RX
IO_L24N_T3_16 G22
IO_25_16 F21
BANK内单端可互换
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U2E
BANK 34
IO_0_34 T3
IO_L1P_T0_34 T1
IO_L1N_T0_34 U1
IO_L2P_T0_34 U2
IO_L2N_T0_34 V2
IO_L3P_T0_DQS_34 R3
IO_L3N_T0_DQS_34 R2
IO_L4P_T0_34 W2
IO_L4N_T0_34 Y2 FPGA_GPIO3
IO_L5P_T0_34 W1
IO_L5N_T0_34 Y1 FPGA_GPIO0
IO_L6P_T0_34 U3
IO_L6N_T0_VREF_34 V3
IO_L7P_T1_34 AA1 FPGA_GPIO1
IO_L7N_T1_34 AB1 FPGA_GPIO2
IO_L8P_T1_34 AB3 FPGA_GPIO5
IO_L8N_T1_34 AB2
IO_L9P_T1_DQS_34 Y3
IO_L9N_T1_DQS_34 AA3 FPGA_GPIO4
IO_L10P_T1_34 AA5 FPGA_GPIO6
IO_L10N_T1_34 AB5 FPGA_GPIO7
IO_L11P_T1_SRCC_34 Y4
IO_L11N_T1_SRCC_34 AA4
IO_L12P_T1_MRCC_34 V4
IO_L12N_T1_MRCC_34 W4
IO_L13P_T2_MRCC_34 R4
IO_L13N_T2_MRCC_34 T4
IO_L14P_T2_SRCC_34 T5
IO_L14N_T2_SRCC_34 U5
IO_L15P_T2_DQS_34 W6
IO_L15N_T2_DQS_34 W5
IO_L16P_T2_34 U6
IO_L16N_T2_34 V5
IO_L17P_T2_34 R6
IO_L17N_T2_34 T6
IO_L18P_T2_34 Y6
IO_L18N_T2_34 AA6
IO_L19P_T3_34 V7
IO_L19N_T3_VREF_34 W7
IO_L20P_T3_34 AB7
IO_L20N_T3_34 AB6
IO_L21P_T3_DQS_34 Y9
IO_L21N_T3_DQS_34 V8
IO_L22P_T3_34 AA8
IO_L22N_T3_34 AB8
IO_L23P_T3_34 Y8
IO_L23N_T3_34 Y7
IO_L24P_T3_34 W9
IO_L24N_T3_34 Y9
IO_25_34 U7
BANK内单端可互换
XC7A75T-1FGG484I
U2F
BANK 35
IO_0_35 F4 FX3_DQ6
IO_L1P_T0_AD4P_35 B1 FX3_DQ1
IO_L1N_T0_AD4N_35 A1 FX3_DQ0
IO_L2P_T0_AD12P_35 C2 FX3_CTL9
IO_L2N_T0_AD12N_35 B2 FX3_CTL2
IO_L3P_T0_DQS_AD5P_35 E1 FX3_DQ2
IO_L3N_T0_DQS_AD5N_35 D1 FX3_DQ5
IO_L4P_T0_35 E2 FX3_DQ15
IO_L4N_T0_35 D2 FX3_CTL12
IO_L5P_T0_AD13P_35 G1 FX3_DQ10
IO_L5N_T0_AD13N_35 F1 FX3_DQ31
IO_L6P_T0_35 F3 FX3_DQ3
IO_L6N_T0_VREF_35 E3 FX3_DQ4
IO_L7P_T1_AD6P_35 K1 FX3_DQ9
IO_L7N_T1_AD6N_35 J1 FX3_CTL4
IO_L8P_T1_AD14P_35 H2 FX3_DQ28
IO_L8N_T1_AD14N_35 G2 FX3_DQ30
IO_L9P_T1_DQS_AD7P_35 K2 FX3_CTL7
IO_L9N_T1_DQS_AD7N_35 J2 FX3_DQ24
IO_L10P_T1_AD15P_35 J5 FX3_CTL1
IO_L10N_T1_AD15N_35 H5 FX3_CTL0
IO_L11P_T1_SRCC_35 H3 FX3_DQ11
IO_L11N_T1_SRCC_35 G3 FX3_DQ8
IO_L12P_T1_MRCC_35 H4 FX3_DQ13
IO_L12N_T1_MRCC_35 G4 FX3_DQ12
IO_L13P_T2_MRCC_35 K4 FX3_INT
IO_L13N_T2_MRCC_35 J4 FX3_DQ7
IO_L14P_T2_SRCC_35 L3 FX3_PCLK
IO_L14N_T2_SRCC_35 K3 FX3_DQ14
IO_L15P_T2_DQS_35 M1 FX3_DQ26
IO_L15N_T2_DQS_35 L1 FX3_CTL3
IO_L16P_T2_35 M3 FX3_CTL11
IO_L16N_T2_35 M2 FX3_DQ23
IO_L17P_T2_35 K6 FX3_CTL6
IO_L17N_T2_35 J6 FX3_CTL5
IO_L18P_T2_35 L5 FX3_DQ17
IO_L18N_T2_35 L4 FX3_CTL8
IO_L19P_T3_35 N4 FX3_DQ27
IO_L19N_T3_VREF_35 N3 AUX_PWR_ON
IO_L20P_T3_35 R1 FX3_GPIO55
IO_L20N_T3_35 P1 FX3_DQ29
IO_L21P_T3_DQS_35 P5 FX3_DQ19
IO_L21N_T3_DQS_35 P4 FX3_DQ22
IO_L22P_T3_35 P2 FX3_GPIO54
IO_L22N_T3_35 N2 SHDN_SW
IO_L23P_T3_35 M6 FX3_DQ21
IO_L23N_T3_35 M5 FX3_DQ16
IO_L24P_T3_35 P6 FX3_DQ18
IO_L24N_T3_35 N5 FX3_DQ25
IO_25_35 L6 FX3_DQ20
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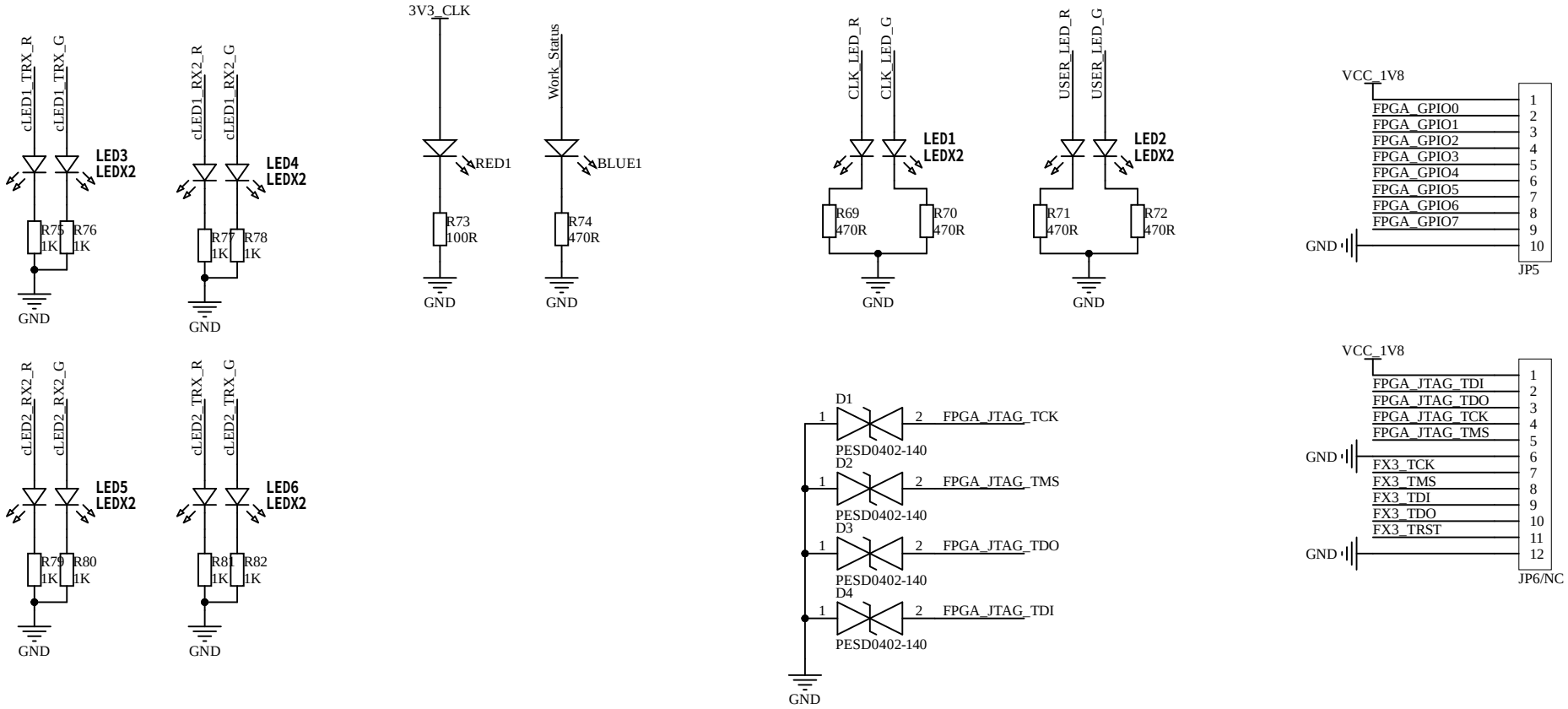

U4B
B12 VSSA
L7 VSSA
L8 VSSA
L9 VSSA
L10 VSSA
L11 VSSA
L12 VSSA
A12 VSSA/TX_EXT_LO
B1 VSSA
B2 VSSA
A4 VSSA
A6 VSSA
C2 VSSA
C7 VSSA
C8 VSSA
C9 VSSA
C10 VSSA
C11 VSSA
C12 VSSA
F3 VSSA
G1 VSSA/RX_EXT_LO
L2 VSSA
L3 VSSA
M4 VSSA
M6 VSSA
H2 VSSA
H3 VSSA
H5 VSSA/SYNC_IN
H6 VSSA
J2 VSSA
K2 VSSA
K12 VSSD
H7 VSSD
H10 VSSD
D12 VSSD
F7 VSSD
F9 VSSD
F11 VSSD
G12 VSSD
VDDA1P3_RX_TX D3
VDDA1P3_RX_RF D2
VDDA1P3_RX_LO E2
VDDA1P3_RX_SYNTH J3
VDDA1P3_RX_VCO_LDO F2
RX_VCO_LDO_OUT G2
VDDA1P1_RX_VCO G3
VDDA1P3_TX_LO_BUFFER E3
VDDA1P3_TX_LO B9
VDDA1P3_TX_SYNTH K3
VDDA1P3_TX_VCO_LDO B10
TX_VCO_LDO_OUT B11
VDDA1P1_TX_VCO A11
VDDA1P3_BB K4
VDD_GPO B8
VDDD1P3_DIG F12
VDD_INTERFACE H12
RBIAS L4
DNC/XTALP M11
DNC M3
DNC A3
1V3_B
1V3_A
C184 1nF
C185 1uF
R61 1R
GND
C186 1nF
C187 1uF
R62 1R
VCC_1V8
L14 L/120R
C188 1uF
R63 14.3K
Power
AD936X
U4C
CLK_40M_RF
C182 100nF
R60 49.9R
C183 100nF
RF_CLK_OUT J6
C4
RF_TXnRX H4
RF_EN_AGC G5
RF_ENABLE G6
B4 GPO_3
B5 GPO_2
B6 GPO_1
B7 GPO_0
RF_CTL_IN0 C5
RF_CTL_IN1 C6
RF_CTL_IN2 D6
RF_CTL_IN3 D5
RF_CTL_OUT0 D4
RF_CTL_OUT1 E4
RF_CTL_OUT2 E5
RF_CTL_OUT3 E6
RF_CTL_OUT4 F6
RF_CTL_OUT5 F5
RF_CTL_OUT6 F4
RF_CTL_OUT7 G4
RF_SPI_DI J4
RF_SPI_CLK J5
RF_SPI_EN K6
RF_SPI_DO L6
RF_RESET K5
XTALN
CLK_OUT
TEST/ENABLE
TXNRX
EN_AGC
ENABLE
GPO_3
GPO_2
GPO_1
GPO_0
CTRL_IN0
CTRL_IN1
CTRL_IN2
CTRL_IN3
CTRL_OUT0
CTRL_OUT1
CTRL_OUT2
CTRL_OUT3
CTRL_OUT4
CTRL_OUT5
CTRL_OUT6
CTRL_OUT7
SPI_DI
SPI_CLK
SPI_EN#
SPI_DO
RESET#
FB_CLK_N G10 RF_FB_CLK_N
FB_CLK_P F10 RF_FB_CLK_P
DATA_CLK_N H11 RF_DATA_CLK_N
DATA_CLK_P G11 RF_DATA_CLK_P
P1_D0/RX_D0_N K11 RF_RX_D0_N
P1_D1/RX_D0_P J12 RF_RX_D0_P
P1_D2/RX_D1_N K10 RF_RX_D1_N
P1_D3/RX_D1_P J11 RF_RX_D1_P
P1_D4/RX_D2_N K9 RF_RX_D2_N
P1_D5/RX_D2_P J10 RF_RX_D2_P
P1_D6/RX_D3_N K8 RF_RX_D3_N
P1_D7/RX_D3_P J9 RF_RX_D3_P
P1_D8/RX_D4_N K7 RF_RX_D4_N
P1_D9/RX_D4_P J8 RF_RX_D4_P
P1_D10/RX_D5_N J7 RF_RX_D5_N
P1_D11/RX_D5_P H8 RF_RX_D5_P
RX_FRAME_P G8 RF_RX_FRAME_P
RX_FRAME_N G7 RF_RX_FRAME_N
P0_D11/TX_D5_P E7 RF_TX_D5_P
P0_D10/TX_D5_N F8 RF_TX_D5_N
P0_D9/TX_D4_P D7 RF_TX_D4_P
P0_D8/TX_D4_N E8 RF_TX_D4_N
P0_D7/TX_D3_P D8 RF_TX_D3_P
P0_D6/TX_D3_N E9 RF_TX_D3_N
P0_D5/TX_D2_P D9 RF_TX_D2_P
P0_D4/TX_D2_N E10 RF_TX_D2_N
P0_D3/TX_D1_P D10 RF_TX_D1_P
P0_D2/TX_D1_N E11 RF_TX_D1_N
P0_D1/TX_D0_P D11 RF_TX_D0_P
P0_D0/TX_D0_N E12 RF_TX_D0_N
TX_FRAME_N H9 RF_TX_FRAME_N
TX_FRAME_P G9 RF_TX_FRAME_P
Interface
AD936X
U4A
TX1A_P M7
TX1A_N M8
TX1B_P M9
TX1B_N M10
RX1A_P M1
RX1A_N M2
RX1B_P H1
RX1B_N J1
RX1C_N L1
RX1C_P K1
TX_MON1 M5
TX_MON2 A5
TX2A_N A7 TX2A_N
TX2A_P A8 TX2A_P
TX2B_N A9
TX2B_P A10
RX2A_P A2 RX2A_P
RX2A_N A1 RX2A_N
RX2B_N F1
RX2B_P E1
RX2C_N D1
RX2C_P C1
AUXDAC1 B3 auxdac1
AUXDAC2 C3
AUXADC L5
1V3_A
1V3_B
GND
RF&IF
AD936X
C189 100nF
C190 100nF
C191 100nF
C192 100nF
C193 100nF
C194 1uF
C195 1uF
C196 100nF
C197 100nF
C198 1uF
C199 100nF
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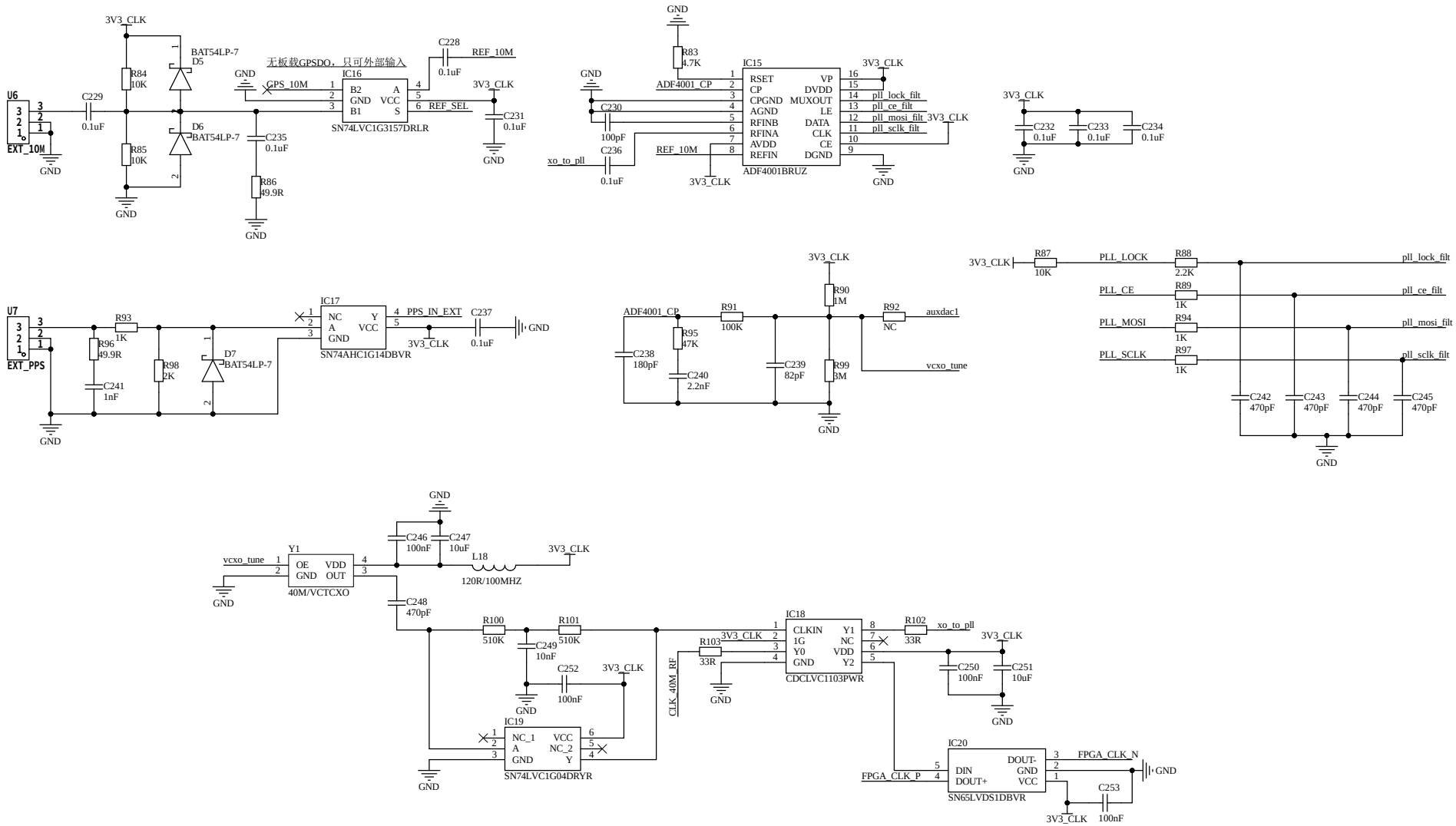
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